Denis Gruszkiewicz

Quantitative Developer Intern | Python, Rust, time series, market data

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github.com/Flyingzaptop | antillia.tech

PROFILE

Engineering student (Robotics and Process Automation, Politechnika Wrocławska; IT engineering, WSB Merito) with a two-year independent research project on gold and financial time series: event-level market data (CME MBO/L3), order-book replay in Rust, feature and label engineering in Python, PyTorch models and strict chronological validation. My results so far are research infrastructure and documented negative findings, not a profitable strategy. Also hold over a year of work as a junior IT administrator.

RELEVANT SKILLS

- Python: NumPy, Pandas, Polars, PyArrow, SciPy, scikit-learn; time-series workflows, preprocessing and feature engineering; PyTorch (LSTM, TCN, attention, custom losses).
- Rust and C++: Rust order-book replay and conservative FIFO execution simulation with zstd-compressed Parquet/DBN data; Rust/PyO3 native extensions; C++/CMake/CTest in a D3D12 performance-analysis project (ARC). Project-based level.
- Quantitative foundation: walk-forward validation, latency and cost stress tests, causality and leakage checks, calibration and stability analysis, controlled baseline comparison.
- Market data infrastructure: asyncio/aiohttp WebSocket collectors for eight exchanges (Binance, OKX, Kraken, Bybit, Coinbase, Bitfinex, Deribit, Hyperliquid), sequence handling, append-only Parquet log with recovery; cTrader API / Protobuf.
- Event-driven and backtesting infrastructure built in own projects; no commercial trading experience. Git/GitHub and GitHub Actions CI with pytest.
- Platform and automation: PostgreSQL, n8n, Django/React application, REST APIs.

EXPERIENCE

Junior IT Administrator | ZOZ w Bolesławcu | 06.2025 - present

- Day-to-day technical support for hospital staff; diagnosing hardware, operating system and application issues.
- Analysed user needs, tested scenarios, reproduced reported errors and fixed applications based on staff needs.
- Developed internal tools for repetitive tasks; worked on separate employee-management, scheduling and laboratory-process applications.

SELECTED PROJECTS

gc-mbo-research | Python, Rust

CME GC event-level data, order-book reconstruction, FIFO execution simulation, bid/ask and latency stress, twenty experiment folders; current README result NO_TRADE (negative results documented).

stargazin and xauusd | Python, PyTorch, Rust/PyO3

Parquet ingestion, features, labels, RetNet-lite / TCN models with attention pooling, offline/online parity manifest, causality tests (github.com/Flyingzaptop).

stargaze-l2-vwap | Python asyncio

Market data collectors, record log with checksums and recovery, VWAP research with frozen paper-only evaluation.

EDUCATION

Politechnika Wrocławska | 10.2022 - present

Robotics and Process Automation; engineering studies in progress, five semesters completed.

WSB Merito we Wrocławiu | 03.2026 - present

Network, systems and IT security engineering; part-time engineering studies, second semester.

LANGUAGES AND AVAILABILITY

Based in Wrocław, Poland (CET, about 3.5 hours from IST). Flexible working hours; fully remote internship suits me. Part-time availability during studies; start date by agreement.